

Pull REquest Charts for an Objective Gauging (PRECOG)

VSCode plugin evaluation

This questionnaire is anonymous and aims to evaluate a VSCode plugin for pull request management. This survey is conducted by students and researchers at the University of Namur, under the supervision of Professors Benoît Vanderose and Xavier Devroey from the SNAIL team (contact: snail.info@unamur.be).

* Obligatoire

Step 0: Presentation of the project

This first step aims to introduce you to the project with a general overview of the different packages to gain an overall idea of the architecture and an overall introduction to the development practices related to pull requests, like naming conventions, etc.

About the spring-framework project: <https://docs.spring.io/spring-framework/reference/overview.html>

1. Do you consent to the recording of your screen and our conversation? *

Acceptez-vous que l'on enregistre votre écran et notre conversation ?

☐ Yes

☐ No

2. How long have you been developing software in a professional context (excluding academic projects)? *

Depuis combien de temps développez-vous du logiciel dans un contexte professionnel (en dehors des projets liés à un cours) ?

- ☐ Less than 6 months
- ☐ Between 6 months and 1 year
- ☐ 1 to 3 years
- ☐ 3 to 5 years
- ☐ 5 to 10 years
- ☐ More than 10 years

3. How would you rate your expertise in code review? *

Comment estimez-vous votre expertise en revue de code ?

Beginner      Expert

Step 1: Ranking of the different PRs

You have 20 minutes to get familiar with the pull requests available at <https://github.com/hraskin/spring-framework/pulls>. After that, please answer the following questions.

4. Please rank the pull requests from easiest (ranked first) to the most difficult (ranked last): *

#6 Add QUERY HTTP method
#3 Prevent NoClassDefFoundError when Jetty Reactive HttpClient is not available
#10 35626: defaultCandidate for scoped proxies
#4 Optimize parameter matching in TableMetaDataContext
#5 SingletonSupplier is not singleton if null is returned on first supplier call
#2 Normalize JAR entry paths to strip BOOT-INF/classes prefix in Spring-Boot fat JARs
#7 Add RSocketServiceMethod support for Kotlin suspending functions
#12 Refactor doGetBeanNamesForType for improved readability and performance
#9 Release data buffer in AbstractCharSequenceDecoder even when String creation fails

5. How confident are you in the accuracy of the ranking? *

Not at all confident	Slightly confident	Neutral	Confident	Very confident
☐	☐	☐	☐	☐

6. Are there any additional comments you want to share?

Step 2: Automated ranking

Please install the PRECOG plugin for VSCode following the instructions available at <https://github.com/snail-unamur/precog-plugin-evaluation/tree/main/precog-plugin>.

In this part, we would like to ask you to focus on the ranking provided by the plugin. **Please do not open a PR in the plugin** right now (we will do that in the next step). Before answering the questions, you may look at the PR on GitHub and/or any other information attached to the PR (SonarQube report, etc.). Please **do not come back to the previous screen** to compare your ranking to the plugin's ranking.

7. To what extent do you agree with the following statements? *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The ranking is representative of the PR difficulties	☐	☐	☐	☐	☐
Difficult PRs are ranked correctly	☐	☐	☐	☐	☐
Easier PRs are ranked correctly	☐	☐	☐	☐	☐

8. For the PRs that are ranked farther from your previous manual ranking, what is your opinion?

9. Are there any additional comments you want to share?

Step 3: Evaluation of the complexity overview

Now you can open the different PRs in the plugin. If you do not remember which PR you selected as the most complex or simple ones, the evaluator can help you.

10. Please specify the PR that you ranked as **the most complex**: *

- ☐ #12 Refactor doGetBeanNamesForType for improved readability and performance
- ☐ #10 35626: defaultCandidate for scoped proxies
- ☐ #9 Release data buffer in AbstractCharSequenceDecoder even when String creation fails
- ☐ #7 Add RSocketServiceMethod support for Kotlin suspending functions
- ☐ #6 Add QUERY HTTP method
- ☐ #5 SingletonSupplier is not singleton if null is returned on first supplier call
- ☐ #4 Optimize parameter matching in TableMetaDataContext
- ☐ #3 Prevent NoClassDefFoundError when Jetty Reactive HttpClient is not available
- ☐ #2 Normalize JAR entry paths to strip BOOT-INF/classes prefix in Spring-Boot fat JARs

11. For the PR above (**the most complex**), to what extent do you agree with the following statement?

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The visualization proposed by the plugin informs you about the complexity of the PR.	☐	☐	☐	☐	☐
The visualization proposed by the plugin is consistent with the complexity of the PR.	☐	☐	☐	☐	☐

12. Please specify the PR that you ranked as **the most simple**: *

- ☐ #12 Refactor doGetBeanNamesForType for improved readability and performance
- ☐ #10 35626: defaultCandidate for scoped proxies
- ☐ #9 Release data buffer in AbstractCharSequenceDecoder even when String creation fails
- ☐ #7 Add RSocketServiceMethod support for Kotlin suspending functions
- ☐ #6 Add QUERY HTTP method
- ☐ #5 SingletonSupplier is not singleton if null is returned on first supplier call
- ☐ #4 Optimize parameter matching in TableMetaDataContext
- ☐ #3 Prevent NoClassDefFoundError when Jetty Reactive HttpClient is not available
- ☐ #2 Normalize JAR entry paths to strip BOOT-INF/classes prefix in Spring-Boot fat JARs

13. For the PR above (**the most simple**), to what extent do you agree with the following statement? *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The visualization proposed by the plugin informs you about the complexity of the PR.	☐	☐	☐	☐	☐
The visualization proposed by the plugin is consistent with the complexity of the PR.	☐	☐	☐	☐	☐

14. For a PR #4 that **the PRECOG plugin** ranked as **complex**, to what extent do you agree with the following statement? *

PR #4 Optimize parameter matching in TableMetaDataContext

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The visualization proposed by the plugin					

The plugin
informs you
about the
complexity of
the PR.

☐☐☐☐☐

The
visualization
proposed by
the plugin is
consistent with
the **complexity**
of the PR.

☐☐☐☐☐

15. For a PR #5 that **the PRECOG plugin** ranked as **simple**, to what extent do you agree with the following statement? *

PR #5 SingletonSupplier is not singleton if null is returned on first supplier call

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

The
visualization
proposed by
the plugin
informs you
about the
complexity of
the PR.

☐☐☐☐☐

The
visualization
proposed by
the plugin is
consistent with
the **complexity**
of the PR.

☐☐☐☐☐

16. Are there any additional comments you want to share?

Step 4: Overall evaluation of the pugin

For the assessment of the product, please fill out the following questionnaire. The questionnaire consists of pairs of contrasting attributes that may apply to the product. The circles between the attributes represent gradations between the opposites. You can express your agreement with the attributes by ticking the circle that most closely reflects your impression.

Example:

attractive ☐ ☐ ☐ ☐ ☐ unattractive

This response would mean that you rate the application as more attractive than unattractive.

Please decide spontaneously. Don't think too long about your decision to make sure that you convey your original impression. Sometimes you may not be completely sure about your agreement with a particular attribute or you may find that the attribute does not apply completely to the particular product. Nevertheless, please make a choice in every line.

It is your personal opinion that counts. Please remember: there is no wrong or right answer!

Afin d'évaluer le produit, veuillez remplir le questionnaire ci-dessous. Il se compose de paires opposées de propriétés que le produit peut avoir. La gradation entre les contraires est représentée par des cercles. En cochant l'un de ces cercles, vous pouvez exprimer votre approbation d'un concept.

Exemple : Attraktif ☐ ☒ ☐ ☐ ☐ Rébarbatif

Cette réponse indique que vous jugez le site plus attraktif que rébarbatif.

Répondez le plus spontanément possible.

Veuillez répondre à toutes les questions, même si vous n'êtes pas sûr de l'évaluation. Il n'y a pas de «bonne» ou de «mauvaise» réponse, seul votre avis compte !

17. 1 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

annoying / *agaçant*

enjoyable / *agréable*

18. 2 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

not understandable /
incompréhensible

understandable /
compréhensible

19. 3 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

creative / *moderne*

dull / *sans fantaisie*

20. 4 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

easy to learn / *ap-
propriation simple*

difficult to learn /
*appropriation com-
pliquée*

21. 5 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

valuable / *apporte
de la valeur*inferior / *peu de va-
leur ajoutée*

22. 6 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

boring / *ennuyeux*exciting / *captivant*

23. 7 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

not interesting / *inin-
téressant*interesting / *intéres-
sant*

24. 8 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

unpredictable / *im-
prévisible*predictable / *prévi-
sible*

25. 9 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

fast / *rapide*slow / *lent*

26. 10 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

inventive / *original*conventional /
conventionnel

27. 11 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

obstructive / *rigide*supportive / *facilitant*

28. 12 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

good / *bien*bad / *médiocre*

29. 13 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

complicated / *com-
pliqué*easy / *simple*

30. 14 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

unlikable / *repous-
sant*pleasing / *attractif*

31. 15 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

usual / *habituel*leading edge
/ *avant-gardiste*

32. 16 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

unpleasant / *désa-
gréable*pleasant / *agréable*

33. 17 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

secure / *sécurisant*not secure / *insécuri-
sant*

34. 18 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

motivating / *stimu-
lant*demotivating / *sopo-
rifique*

35. 19 *

1	2	3	4	5	6	7
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meets expectations
/ répond aux at-
tentes

does not meet ex-
pectations / ne ré-
pond pas aux at-
tentes

36. 20 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

inefficient / inefficace

efficient / efficace

37. 21 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

clear / clair

confusing / dérou-
tant

38. 22 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

impractical / non
pragmatique

practical / pragma-
tique

39. 23 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

organized / sobre

cluttered / surchargé

40. 24 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

attractive / attrayant

unattractive / rébar-
batif

41. 25 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

friendly / sympa-
thique

unfriendly / inamical

42. 26 *

1	2	3	4	5	6	7
---	---	---	---	---	---	---

conservative /
conservateur

innovative / *innovant*

Step 4: Closing questionnaire

Here are some final questions about your overall experience with the plugin.

43. How would you rate your overall experience with the plugin? *

Very poor ☆ ☆ ☆ ☆ ☆ Excellent

44. How easy was the plugin to understand? *

Very Difficult 😞 😞 😞 😞 😞 Very Easy

45. To what extent does the plugin accurately represent the complexity of a PR? *

Not at all 🏆 🏆 🏆 🏆 🏆 Completely

46. To what extent do you think the plugin adds real value when selecting a PR? *

Not at all 🏆 🏆 🏆 🏆 🏆 A great deal

47. How likely are you to use the plugin regularly in your work? *

Not at all likely 👍 👍 👍 👍 👍 Extremely likely

48. In your opinion, what is the main strength of the plugin?

49. In your opinion, what is the main weakness of the plugin?

50. Did you find any metrics **in the radar** unnecessary or not relevant?

☐ Cyclomatic complexity

☐ Instability

☐ Lack of coherence

51. In your opinion, which metric(s) should be **added to the radar**?

52. Did you find any metrics **outside of the radar** unnecessary or not relevant?

- ☐ Files count
- ☐ Lines count
- ☐ New coverage

53. In your opinion, which metric(s) should be **added outside of the radar**?

54. If you could change one thing about the plugin, what would it be?

55. Are there any additional comments you want to share?

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